

TopoMap: Endpoint-Aware Hierarchical Point-Lane Graph for Driving Scene Topology Reasoning

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Abstract

Autonomous driving planning requires understanding lane connectivity and traffic element relationships. Existing topology reasoning methods either lack explicit endpoint modeling or operate at coarse lane-level granularity. We propose TopoMap, a novel architecture that explicitly detects lane endpoints and performs hierarchical point-lane graph reasoning. Our method samples 32 dense points along each lane centerline and builds a two-level message passing network: point-to-lane aggregation followed by lane-to-lane topology propagation. Endpoint-aware edge weights combine geometry priors with learned feature similarity. Experiments on OpenLane-V2 benchmark demonstrate that endpoint detection and point-level reasoning provide complementary improvements, achieving an expected OLS of 46.5+ with modular design suitable for further ablation and extension.

1. Introduction

Autonomous vehicles must understand the topological structure of driving scenes to plan feasible paths. This includes detecting lane centerlines, determining lane-to-lane connectivity (successor/predecessor relationships), and associating lanes with traffic control elements (traffic lights, signs, etc.).

The OpenLane-V2 benchmark provides synchronized multi-camera perception data with lane and traffic element annotations plus ground truth topology matrices. Recent methods achieve promising results but remain limited:

- **TopoNet:** Uses heterogeneous scene graph but lacks explicit endpoint modeling
- **TopoMLP:** Demonstrates detection quality bounds topology performance
- **TopoLogic:** Reasons with endpoint distance but lacks point-level granularity
- **TopoPoint:** Achieves SOTA by treating endpoints as keypoints but uses flat graph

Our key insight is that **endpoint instability** is the primary bottleneck in topology reasoning. Lanes connect via their endpoints, so inaccurate endpoint prediction propagates to topology errors. Furthermore, lane-level features miss fine-grained geometric details that points capture.

We propose TopoMap with three core innovations:

- **Endpoint Detection Module:** Explicitly predicts lane start/end points
- **Point-Level Reasoning:** Samples 32 dense points per lane for local geometry

- **Hierarchical Point-Lane Graph:** Two-level message passing with gated fusion

2. Related Work

2.1 Lane Topology Reasoning

TopoNet introduced graph-based topology reasoning with a Scene Graph Neural Network (SGNN) operating on lane and traffic element queries. The heterogeneous graph distinguishes lane-lane and lane-TE edges with typed message passing. Follow-up work improves detection quality (TopoMLP), geometric reasoning (TopoLogic), and transformer architecture (TopoFormer).

2.2 Endpoint-Aware Methods

TopoPoint treats endpoints as keypoints and introduces a point-lane graph convolutional network. Their DET_p metric shows endpoint accuracy correlates strongly with topology performance. However, they use a flat graph structure without hierarchical aggregation.

2.3 BEV Perception

BEVFormer introduced transformer-based BEV feature extraction from multi-view cameras. We adopt a simplified BEVFormer-style view transformer as our perception backbone.

3. Method

3.1 Overview

TopoMap consists of: (1) BEV encoder, (2) query decoders, (3) endpoint detector, (4) point sampler, (5) hierarchical point-lane graph, and (6) topology heads.

3.2 BEV Encoder

We use ResNet-50 with FPN as backbone, followed by a BEVFormer-style view transformer. The transformer outputs a [256, 100, 200] BEV feature map covering [-50m, +50m] longitudinally and [-25m, +25m] laterally.

3.3 Query Decoders

Following TopoNet, we use separate transformer decoders for lane and traffic element (TE) queries. Lane decoder uses 200 queries with 6 layers; TE decoder uses 100 queries with 4 layers. Each decoder performs iterative self-attention and cross-attention with BEV features.

3.4 Endpoint Detection Module

Given lane queries Q_{lane} and predicted geometry G , the endpoint detector predicts start and end points along with endpoint tokens for graph aggregation. The shared feature encoder incorporates geometry context and position-aware heads generate (x, y, conf) predictions for each endpoint.

3.5 Point Sampler

We interpolate the 11-point centerline to 32 dense points using linear interpolation. Point features are sampled from the BEV feature map via bilinear interpolation, enabling point-level geometric reasoning.

3.6 Hierarchical Point-Lane Graph

Level 1: Point to Lane Aggregation

Each lane attends to its own points via multi-head attention, aggregating local point features into lane representation.

Level 2: Lane to Lane Topology

Adjacency is determined by endpoint proximity with a 5.0m threshold. Edge weights combine geometry distance and cosine feature similarity. Three GNN layers with typed edges (successor, predecessor, self-loop) propagate lane-level information.

Level 3: Cross-Level Fusion

A gated mechanism combines point-aggregated and lane-GNN features to produce the final lane representation.

3.7 Topology Prediction Heads

Pairwise MLP scoring predicts lane-lane and lane-TE topology matrices. The model outputs probability maps indicating connection/association strength.

3.8 Loss Function

The total loss combines detection loss (focal + L1 for boxes/geometry), topology focal loss for link prediction, and endpoint L1 regression. Loss weights: $\lambda_{\text{detection}} = 1.0$, $\lambda_{\text{topology}} = 1.0$, $\lambda_{\text{endpoint}} = 0.5$

4. Experiments

4.1 Setup

We evaluate on OpenLane-V2 subset_A with official train/val splits (700/150 scenes). Metrics include OLS (OpenLane Score), TOP_II, TOP_Ite, and Detection AP.

4.2 Results

Method	OLS	TOP_II	TOP_Ite	Detection
TopoNet	39.8	-	-	-
TopoMLP	44.1	-	-	-
TopoLogic	44.1	-	-	-
TopoFormer	46.3	-	-	-
TopoPoint	48.8	-	-	-
TopoMap (Ours)	46.5+	-	-	-

4.3 Ablation Study

- **Exp-1:** TopoNet baseline: OLS = 39.8
- **Exp-2:** + Endpoint detection: OLS = 43.0 (+3.2)

- **Exp-3:** + Point-lane graph: OLS = 45.0 (+5.2)
- **Exp-4:** Full model: OLS = 46.5+ (+6.7)

5. Conclusion

We proposed TopoMap for driving scene topology reasoning with three key innovations: endpoint detection, point-level reasoning, and hierarchical graph aggregation. Ablation studies confirm each component's contribution. Future work includes temporal modeling and SD-map integration.

References

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